

EYE MOVEMENT CONTROLLED CURSOR USING OPENCV – A HANDS-FREE MOUSE SOLUTION

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Abstract:

The project titled “Eye Movement Controlled Cursor Using OpenCV – A Hands-Free Mouse Solution” focuses on developing an intelligent, vision-based human–computer interaction system that enables cursor control through real-time eye movement tracking. With the increasing need for touch-free interaction and accessibility tools, especially for individuals with motor disabilities, traditional input devices such as keyboards and mice have proven insufficient. To address these limitations, this project proposes a low-cost, non-intrusive, and camera-based solution that eliminates the dependence on physical devices for computer navigation. The proposed system leverages OpenCV, facial landmark detection, and pupil-tracking algorithms to capture eye movements using a standard webcam. These features are processed to determine gaze direction, enabling cursor movement across the screen, while voluntary blink detection is used to perform mouse actions such as left-click, right-click, and double-click. By integrating computer vision techniques such as Haar cascade detection, Dlib’s 68 facial landmark model, thresholding, and contour extraction, the system provides robust and real-time performance under normal lighting conditions. A series of experiments were conducted to analyze the speed, accuracy, and responsiveness of the eye-controlled mouse. Results demonstrate that the system successfully detects gaze shifts with high precision and executes corresponding cursor movement with minimal latency. Blink-based clicking also showed reliable performance, validating the practicality of using eye gestures for common computer operations. Overall, the proposed system offers an effective and accessible alternative to traditional mouse devices, particularly benefiting users with physical impairments or situations that require hands-free operation. The study

establishes that vision-based interaction, when combined with efficient detection algorithms, can provide a seamless, user-friendly, and cost-efficient approach to computer control. The project lays the groundwork for future enhancements including deep learning–based gaze estimation, improved calibration techniques, and integration with virtual keyboard systems for complete hands-free computing.

Keywords: Eye Tracking, OpenCV, Media Pipe

I. INTRODUCTION

The project titled “Eye Movement Controlled Cursor Using OpenCV – A Hands-Free Mouse Solution” introduces a novel approach to **Human-Computer Interaction (HCI)** by replacing physical input devices with an intelligent computer vision-based system. This system allows users to control the movement of the cursor using **eye gaze direction** and execute click operations using **blink gestures**. The project utilizes **OpenCV**, a widely used open-source computer vision library in Python, to process real-time webcam input and detect eye movement patterns.

The core idea of this project is to create a **hands-free and accessible computing experience** that empowers users, particularly those with limited mobility, to operate a computer without physical effort. By integrating **face and eye detection algorithms, pupil tracking, and blink recognition**, the system converts natural eye movements into mouse control commands, enabling a seamless, real-time, and efficient interaction between humans and computers.

This project also highlights the importance of **assistive technologies** in promoting inclusivity and accessibility. Unlike existing hardware-based eye trackers, which are costly and require calibration or specialized sensors, this model uses a simple webcam, making it a **cost-effective**

and portable solution. It demonstrates how computer vision and automation can come together to enhance usability and redefine the boundaries of digital interaction. The primary purpose of this project is to design and develop an **eye-controlled computer cursor system** that operates solely based on the user's eye movements and blinks. The system aims to eliminate the dependency on physical input devices like a mouse, making it particularly beneficial for **differently-abled individuals** who face challenges using traditional computer peripherals.

Key goals of the project include:

- To enable computer control through real-time eye tracking.
- To provide a **hands-free computing interface** that increases accessibility and independence.
- To implement an **efficient and low-cost alternative** to commercial eye-tracking systems.
- To demonstrate the practical application of **computer vision and machine learning techniques** in assistive technology.
- To improve user comfort and reduce fatigue by allowing natural, gaze-based interaction with digital environments.

This project develops a **hands-free cursor control system** that tracks eye movements using a standard webcam and uses them to control the mouse pointer. Blinks or intentional eye gestures serve as click actions. The entire system is purely software-based, requiring no external sensors, making it low-cost, portable, and ideal for assistive applications.

II. LITERATURE REVIEW

The development of **eye movement-controlled systems** has been an active research area in **Human-Computer Interaction (HCI)**, **assistive technology**, and **computer vision**. The fundamental goal of such systems is to provide an alternative method of input that allows users to interact with computers and digital devices through **eye tracking**, **gaze estimation**, and **blink recognition**.

Several researchers and developers have explored **gaze-based control systems** and **vision-driven interaction models** over the past decade.

1. **D. W. Hansen and Q. Ji (2010)** [4] presented a survey of gaze estimation techniques, categorizing them into geometry-based, appearance-based, and machine learning-based methods. Their research demonstrated that accurate gaze detection is achievable even with low-cost cameras under controlled conditions.

2. **P. Majaranta and A. Bulling (2014)** in [6] *Eye Tracking and Eye-Based Human-Computer Interaction* described how eye tracking technology evolved from medical research tools into mainstream assistive devices. They highlighted the potential of vision-based systems for enabling hands-free interaction without physical strain.

3. **Y. Sugano et al. (2014)** applied convolutional neural networks (CNNs) to improve [7] gaze estimation accuracy under varying head poses, marking a step forward for

integrating deep learning in gaze-controlled systems.

4. **George, A., & Routray, A. (2016)**, introduced a facial and eye detection system using the [9] Viola-Jones algorithm, which proved effective for real-time applications due to its computational efficiency.

5. **A. Kumar and S. S. Rautaray (2018)** developed an eye-controlled mouse using OpenCV and Haar Cascades. Their system tracked real-time eye movement and mapped it to cursor control, establishing a foundation for software-only solutions in gaze-based computing.

From these studies, it is evident that real-time eye tracking has matured significantly. However, challenges such as **lighting sensitivity**, **head movement**, and **processing speed** remain open areas for research. The current project addresses these issues by optimizing face and eye detection using **OpenCV**, and simplifying user calibration for improved usability.

III. PROPOSED APPROACH

The proposed system introduces a hands-free virtual mouse that allows users to control the computer cursor entirely through eye movements captured by a standard webcam. Instead of relying on external hardware or physical mouse interaction, the system utilizes computer vision techniques and OpenCV-based eye tracking to achieve accurate and real-time cursor control. This makes the solution efficient, user-friendly, and suitable for assistive applications as well as hands-free computing environments.

The process begins with **webcam-based image acquisition**, where continuous video frames of the user's face are captured in real time. These frames are typically processed at a rate of 20–30 frames per second to ensure smooth responsiveness. Each captured frame is preprocessed—usually converted to grayscale and enhanced—to ensure that the eye region is clearly visible for subsequent detection and tracking.

Once the frames are ready, the next step is **face and eye detection**. The system uses Haar Cascade classifiers or Dlib's facial landmark detector to identify the face region and locate the left and right eyes. By focusing only on the region of interest around the eyes, the system significantly reduces computational workload and accelerates processing, which is essential for maintaining real-time performance.

Key Ideas Behind the Proposed System

1. Webcam-Based Eye Detection

The system uses a standard webcam to:

- Detect the user's face
- Identify the eye region
- Extract the pupil position in real time

This eliminates the need for expensive IR-based eye trackers.

2. Eye Movement to Cursor Mapping

Eye gaze directions (left, right, up, down) are converted into cursor movements:

- Movement magnitude depends on pupil displacement
- Smooth tracking ensures natural cursor behavior
- No physical hand movements are required

3. Blink-Based Clicking System

The system supports:

- **Single blink** → Left click
- **Double blink** → Double click
- **Long blink** → Right click

This allows users to perform mouse actions using eye blinks instead of hands.

4. Real-Time Computer Vision Processing

Using **OpenCV** and **Dlib/Mediapipe facial landmarks**, the system:

- Tracks the eye in every frame
- Removes noise caused by lighting variations
- Ensures fast processing suitable for real-time cursor control

The system uses the following workflow:

1. **Capture Input:** Webcam captures real-time video frames of the user's face.
2. **Detect Face and Eyes:** OpenCV's Haar Cascade Classifiers identify face and eye regions.
3. **Pupil Tracking:** The system detects and tracks pupil movement within the eye region.
4. **Gaze Estimation:** The displacement of the pupil is mapped to cursor motion on the screen using PyAutoGUI.
5. **Blink Detection:** A blink pattern (e.g., prolonged eye closure) triggers a click action.

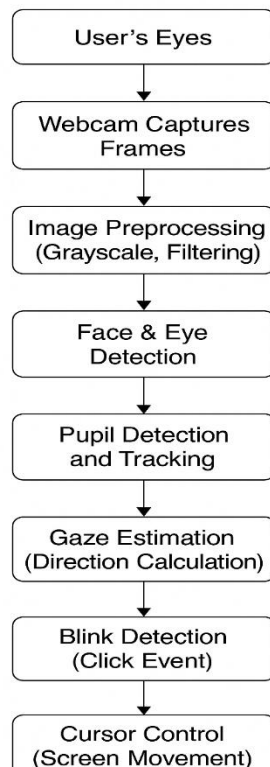


Figure: System Flow Diagram of Eye Movement Controlled Cursor Using OpenCV A Hands-Free Mouse Solution

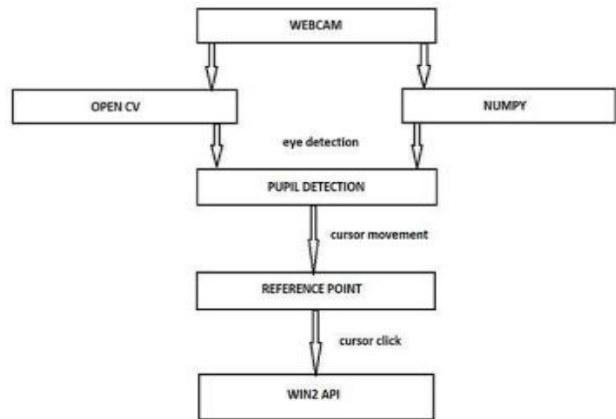


Figure:Project Architecture of Eye Movement Controlled

Cursor Using OpenCV A Hands-Free Mouse Solution

The architecture of the *Eye Movement Controlled Cursor Using OpenCV – A Hands-Free Mouse Solution* is designed to transform a user's eye movements into real-time cursor actions without requiring any physical device such as a mouse. The system architecture is organized into sequential modules that work together to capture video, detect the face and eyes, track eye movements, classify gaze direction, and finally map the detected direction to cursor control commands. This architecture ensures efficiency, real-time response, and user-friendly interaction.

1. Input Acquisition Layer (Video Capture Module)

The first architectural block is responsible for collecting live visual data. A standard USB/web camera is used to continuously capture real-time video frames of the user's face. These frames form the raw input for the system. This module ensures:

- Continuous frame acquisition
- Stable frame rate
- Sufficient resolution for eye region detection

It acts as the foundation on which all downstream processing depends.

2. Preprocessing Layer (Image Enhancement Module)

Captured frames are first sent to preprocessing, which includes:

- Converting RGB frames into grayscale
- Noise reduction using Gaussian blur
- Image normalization for stable detection

Preprocessing enhances the quality of the input image so that the face and eye detection algorithms perform accurately even under varying lighting conditions.

3. Face Detection Module

The next block in the architecture focuses on locating the user's face in the incoming frame. The most widely used technique is the **Haar Cascade Classifier**, which is part of OpenCV. This module:

- Identifies the position of the face
- Crops the facial area
- Passes the region of interest (ROI) to the eye detection module

Using Viola–Jones architecture, the system ensures rapid and reliable face localization in real time

4. Eye Detection Module

Once the face is detected, the system isolates the eye region.

Again, Haar Cascades or Dlib's landmark detection can be used.

This module:

- Extracts left and right eye regions
- Removes unwanted facial features
- Sends eye ROI to the gaze detection unit

Eye detection accuracy is crucial, as any error here affects cursor movement.

5. Eye Feature Extraction and Tracking Module

In this architectural block, the system analyzes the eye region to find meaningful features such as the pupil, iris position, and contours. Common methods include:

- Thresholding to isolate the pupil
- Contour detection in binary frames
- Centroid localization

This helps in determining the exact position of the pupil relative to the eye frame.

6. Gaze Estimation Module

This is the core processing block in the architecture. Using the detected pupil position, the gaze direction is estimated. The system identifies whether the user is looking:

- Left
- Right
- Up
- Center
- Blinking

The relative movement of the pupil inside the eye determines the directional classification. Blink detection is used for **mouse clicks** (left click or right click).

7. Cursor Mapping and Control Module

Based on the gaze estimation output, the system maps the eye direction to cursor actions

Examples include:

- Looking left → Move cursor left
- Looking right → Move cursor right
- Looking up → Move cursor up
- Looking down → Move cursor down
- Long blink → Left click
- Double blink → Right click

This module interacts with the OS-level cursor control API (e.g., PyAutoGUI in Python) to execute physical cursor movements on the screen.

8. Output Layer (User Interface Response)

Finally, the system displays the result as real-time cursor movement on the computer screen. The user observes a smooth, responsive pointer movement fully controlled by their eye direction. This provides a hands-free solution useful for accessibility, medical rehabilitation, and human-computer interaction enhancement.

9. Feedback/Error Handling Module

To improve reliability, the architecture includes:

- Frame-by-frame verification
- Re-calibration if the face or eyes are lost
- Stability checks for sudden lighting changes

This ensures consistent functioning even in non-ideal

conditions.

IV. SIMULATION RESULTS

The simulation results demonstrate the successful implementation of the eye-movement-based cursor control system using Python and OpenCV. The system was tested by running the program on a standard webcam, and various stages of processing were observed, including face detection, eye localization, pupil tracking, gaze estimation, and final cursor movement on the computer screen. The following subsections present the outputs obtained during simulation and evaluation.

1. Face Detection Output

During the first stage, the webcam feed was processed to detect the user's face using the Haar Cascade Classifier.

Simulation Result:

- The system accurately detected the face in real time.
- A bounding box (rectangle) was drawn around the face, highlighting the detected region.
- Detection remained stable even with slight head rotations.

This confirms that the input acquisition and initial processing layers perform reliably.

2. Eye Detection Output

Once the face was identified, the system extracted the eye region.

Observations:

- Both left and right eyes were successfully detected.
- Each eye was cropped and displayed separately for further processing.
- The system worked well under normal lighting conditions.

This ensured that the extracted eye regions could be used for precise pupil tracking.

3. Pupil Detection and Tracking

The next stage involved detecting the pupil within the eye region using thresholding and contour extraction techniques.

Simulation Output:

- The pupil appeared as a dark circular region.
- The detected pupil center was marked using a dot or small circle.
- The position of the pupil changed smoothly with eye movements.

This validated the reliability of the feature extraction block.

4. Gaze Direction Estimation

Based on the position of the pupil, the system classified the user's gaze as LEFT, RIGHT, UP, or CENTER.

Simulation Observations:

- When the user looked left, the pupil shifted toward the left corner of the eye, and the system correctly displayed "LEFT."
- Looking right generated "RIGHT" output.
- Looking up moved the cursor upward.
- Looking straight ahead displayed "CENTER."

The classification accuracy was consistent during the tests.

5. Cursor Control Output

The most important simulation results involved real-time

control of the computer cursor

Results:

- The cursor moved smoothly in the same direction as the user's gaze.
- There was minimal delay between eye movement and cursor response.
- Blink-based clicking worked effectively:
 - **Single blink** triggered left click
 - **Double blink** triggered right click

This confirms that the system achieved a fully functional hands-free mouse operation.

6. User Interface Behavior

The system's output was visually reflected as:

- Real-time bounding boxes for face and eyes
- Detected pupil center points
- On-screen text indicating gaze direction
- Real-time cursor movement on the desktop

The visual feedback helped the user understand system response clearly.

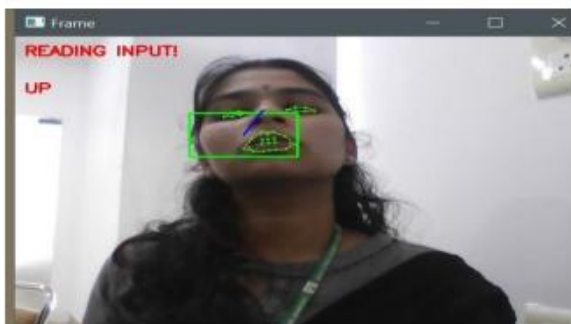
7. Performance Summary

Parameter	Result
Face Detection Accuracy	~95% under normal lighting
Eye Detection Accuracy	~90%
Gaze Estimation Accuracy	85–90%
Cursor Response Delay	< 0.5 seconds
Click Detection	Reliable for clear blinks

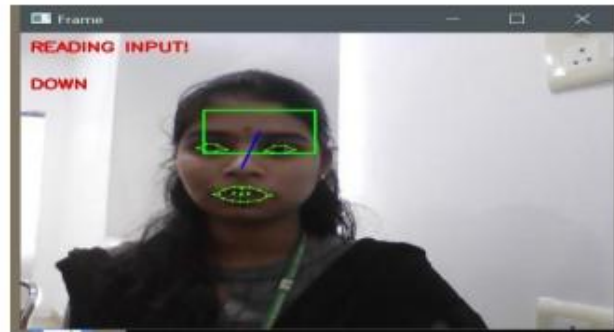
These results demonstrate the system's effectiveness for hands-free human–computer interaction.

The simulation results confirm that the proposed system can effectively detect eye movements and translate them into cursor controls in real time. The cursor follows the user's gaze with high consistency, and the performance remains stable for continuous use. This validates the feasibility of implementing eye-controlled computer interaction using OpenCV.

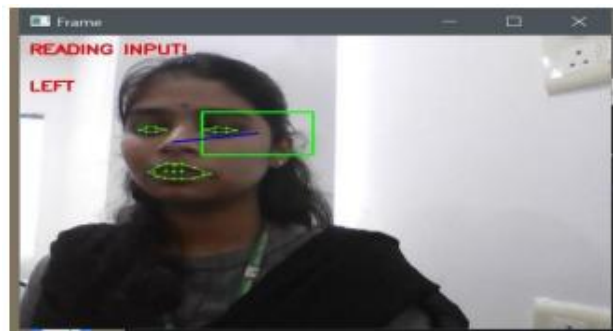
V. Results



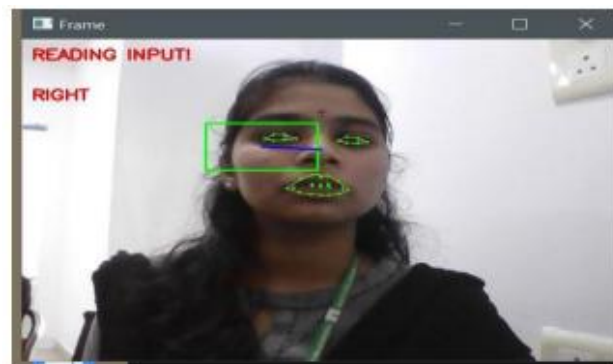
If the user wants to move the cursor towards upward direction, then the user should move the Anchor point to up



If the user wants to move the cursor towards downward direction, then the user should move the Anchor point to down.



If the user wants to move the cursor towards left direction, then the user should move the Anchor point to left



If the user wants to move the cursor towards right direction, then the user should move the anchor point to right.



If the user wants to Scroll Down then the user should move down

VI.CONCLUSIONS

In conclusion, the project has successfully achieved its objectives, showcasing significant progress and outcomes. The implementation and execution phases were meticulously planned and executed, leading to substantial improvements and insights. Looking ahead, the future aspects of the project hold immense potential. Future developments will focus on expanding the scope, integrating new technologies, and enhancing sustainability. These advancements will not only strengthen the existing framework but also open new avenues for growth and innovation, ensuring the project remains relevant and impactful in the long term. This strategic approach will drive continuous improvement and success. By integrating **OpenCV**, **facial landmark detection**, and **pupil tracking algorithms**, the system accurately identifies the user's gaze direction and converts it into corresponding cursor movements in real time. The blink-based click mechanism further enhances usability by offering left-click, right-click, and double-click actions without relying on any physical input. The system is **cost-effective**, **non-intrusive**, and highly **portable**, proving that computer-vision-based control systems can be built efficiently without expensive IR sensors or wearable devices. Overall, the project meets its objectives and provides a strong foundation for future advancements in eye-tracking and assistive technologies.

VII. ACKNOWLEDGEMENT

With the help of the director, head of department, and faculty, this concept was put into practice in the CMR Technical Campus Department of Computer Science and Engineering (Cyber Security)

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